Wavelets and Autoencoders Are All You Need

Anonymous Author(s)
Affiliation
Address
email

Abstract

N-BEATS [**?**] broke new ground in univariate time series forecasting by using fixed basis functions in a neural basis expansion in which the network learns coefficients for a frozen basis function and then expands those coefficients into the desired forecast and backcast lengths through a linear transformation. Their innovation came at an extreme cost, as a typical N-BEATS model can range from 25M–44M parameters. In this paper we extend the idea of solving for coefficients of a frozen basis function but, instead of relying on a global Fourier function, we use wavelets, which provide localization in that they can identify and locate components of the global signal — a key requirement in time series analysis algorithms. Wavelets have been used extensively in signal analysis for feature extraction, denoising, and multi-resolution analysis. Likewise, autoencoders are useful constructs for identifying key characteristics of a signal while filtering out non-essential transient components, and are useful in domains such as denoising signals and stock trading. We introduce a variety of novel block types that use a combination of discrete wavelet transforms (DWT) and autoencoders. Our novel architecture improvements reduce parameter counts by 60–95% over N-BEATS while surpassing or equaling prior published results without ensembling. In addition, we introduce a family of combined TrendWavelet block types that combine a low-pass trend basis with a high-pass wavelet basis to produce <1M parameter models that score within 0.5% sMAPE of 19M–43M parameter baseline N-BEATS models, representing a 12–80× compression with little to no loss of accuracy. Furthermore, we show that these new block constructs are transferable to other architectures like NHiTS and Transformers with equally impressive reductions in parameter counts and competitive accuracies.

1 Introduction and Prior Work

1.1 Introduction

Time series forecasting and signal analysis are two names for the same thing, even though practitioners in each discipline use their own terminologies: moving average vs. finite impulse response (FIR) filter, autoregressive filter vs. infinite impulse response (IIR) filter, seasonal decomposition vs. band-pass filter of seasonal frequencies. A time series dataset (sales figures, temperature, stock prices) is just a discrete set of sampled data. Likewise, all deep learning problems can be reconsidered as signal analysis problems. For example, image recognition in deep learning is often performed by sampling an image and passing the samples through a convolutional filter, similar in principle to what one might do in Photoshop when applying a sharpening filter. Deep learning, then, can be seen as a type of probabilistic signal analysis. Why, then, don't we use more techniques developed in signal analysis in machine learning?

Signal analysis tools like the Discrete Fourier Transform and the Discrete Wavelet Transform were developed as techniques used in signal processing, migrated into forecasting, and are trickling into

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

machine learning. N-BEATS was a pioneer of sorts in bringing some traditional signal analysis tools into AI with its use of basis expansion blocks. The N-BEATS Seasonality and Trend blocks are literally signal-decomposition operators applied inside a neural network. The novelty was using the neural network to learn which decomposition coefficients best reconstruct the observed signal.

1.2 Wavelets

Fourier analysis decomposes a signal into a sum of sine and cosine waves of fixed frequency. It is a useful tool for analyzing stationary signals whose frequency content does not change over time. Real-world signals routinely break that mold, however. For example, a stock price may trend smoothly following the normal sales cycle and then spike abruptly because of a news article, a comment made by a prominent person in the news, or when a large hedge fund liquidates a considerable position. Fourier bases represent these transient features poorly and tend to smear localized events across the other frequency components of the signal.

Wavelets address this problem by providing basis functions that are localized in both time and frequency. Where a sine wave oscillates forever, a wavelet is a brief oscillatory burst with exponentially decaying tails and a zero mean. They oscillate in a particular pattern defined by their mother wavelet and then decay to zero.

The mother wavelet $\psi(t)$ is scaled and translated to produce a family of basis functions:

$$\psi_{j,k}(t) = 2^{j/2}\,\psi(2^j t - k)$$

where j controls scale (frequency resolution) and k controls translation (time position). Short, high-frequency wavelets have narrow time resolution; long, low-frequency wavelets have narrow frequency resolution. This makes wavelets substantially more effective for real-world time series analysis, where variance and frequency content change over time.

A discrete wavelet transform (DWT) brings the continuous wavelet into the digital domain. DWTs compute the inner product of a signal with a basis family, producing a set of orthonormal coefficients that can then be used to reconstruct the original signal [**??**].

Wavelets have been used sparsely in machine learning. WaveTS [**?**] learns a filter-bank specialized to fit non-stationary signals by learning the basis function through decomposing its n-th order derivatives and then learning an affine transformation that lets the network suppress and cross-mix frequencies in training.

MultiWave [**?**] is a model-agnostic preprocessor around any neural network architecture. It takes the concept of machine-learning/signal-processing equivalency to heart by preprocessing input data with a DWT, grouping the outputs by frequency, and then feeding the groups into any neural network (Transformer, FFN, CNN, etc.), feeding different groups into different networks. The networks' outputs are combined and an inverse DWT is applied to obtain the prediction.

W-Transformers [**?**] uses a similar approach by preprocessing time-series data with a discrete overlapping wavelet transform and then feeding the results into a collection of transformer models to analyze the decomposed signals independently.

Most directly related, **?** apply DWT preprocessing to N-BEATS itself, training separate Generic stacks per sub-band; our work differs by embedding the wavelet basis inside each block rather than as an external preprocessor.

1.3 Autoencoders

An autoencoder is a neural network trained to reproduce its input. It has an encoder that maps an input x into a smaller-dimensional latent code z, and a decoder that maps z back into a reconstructed version of the original input $\hat{x}$:

```
x -> encoder -> z -> decoder -> x_hat
loss = reconstruction_error(x, x_hat) + optional_regularization
```

The premise is that the model is constrained by the reduced dimensionality of the latent space. Therefore, since the decoder network does not have access to the original signal, a perfect copy is difficult. Thus the network is pushed to learn how to reconstruct the original signal from the information it can

extract from the latent representation. The classic paper of Hinton and Salakhutdinov [**?**] showed that autoencoders generalize PCA to non-linear domains. Autoencoders are also used widely for self-supervised pretraining, generative modeling, and image denoising and generation [**?**].

2 Model Description

N-BEATS [**?**] analyzes a time series through a sequential stack of doubly residual blocks, each contributing a backcast $\hat{\mathbf{x}}_\ell$ and forecast $\hat{\mathbf{y}}_\ell$ via a learned linear mapping from a shared 4-layer fully connected backbone:

$$\mathbf{h}_\ell = f_{\text{bb}}(\mathbf{x}_\ell^{\text{in}}), \quad \hat{\mathbf{x}}_\ell = g_b(\boldsymbol{\theta}^b), \quad \hat{\mathbf{y}}_\ell = g_f(\boldsymbol{\theta}^f)$$

where $\mathbf{x}_\ell^{\text{in}} = \mathbf{x} - \sum_{k<\ell} \hat{\mathbf{x}}_k$ is the residual backcast input to block ℓ, and g_b, g_f are basis expansion generators (defined explicitly in Section 2.3).

All N-BEATS blocks accept a residual backcast (or input if first block) $\mathbf{x} \in \mathbb{R}^L$ and return $(\hat{\mathbf{x}}, \hat{\mathbf{y}}) \in \mathbb{R}^L \times \mathbb{R}^H$.

2.1 Backbone Blocks

2.1.1 RootBlock

The legacy RootBlock is a 4-layer, uniform-width FFN with ReLU activation σ:

$$\mathbf{h} = \sigma(\mathbf{W}_4\, \sigma(\mathbf{W}_3\, \sigma(\mathbf{W}_2\, \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3) + \mathbf{b}_4)\,, \quad \mathbf{W}_i \in \mathbb{R}^{U \times U}$$

No bottleneck is imposed; all layers share trunk width U.

2.1.2 AERootBlock

The AERootBlock is a drop-in replacement for the legacy RootBlock backbone. It introduces an hourglass-shaped autoencoder that routes all information through a low-dimensional latent bottleneck $z \in \mathbb{R}^d$:

$$\mathbf{h} = f_{\text{dec}}(f_{\text{enc}}(\mathbf{x})), \quad f_{\text{enc}} : \mathbb{R}^L \to \mathbb{R}^{U/2} \to \mathbb{R}^d, \quad f_{\text{dec}} : \mathbb{R}^d \to \mathbb{R}^{U/2} \to \mathbb{R}^U$$

The bottleneck acts as a signal denoiser, regularizer, and feature extractor. The decoder must reconstruct both output paths from z, forcing the backbone to discard irrelevant information and retain the most informative features.

The parameter count of the backbone reduces to $O(UL/2 + Ud/2 + Ud/2 + U^2/2)$. At $U = 512$, $L = 480$, $d = 16$, this is approximately 147K — a 5.5× reduction relative to RootBlock at equivalent U. The interface to any downstream projection head remains identical: both backbones produce $h \in \mathbb{R}^U$.

2.1.3 AERootBlockLG

The LG variant of AERootBlock introduces a learnable sigmoid vector and scales the interior latent representation element-wise:

$$z_{\text{gated}} = \sigma(\gamma) \odot z$$

Because $\sigma(\gamma_i)$ saturates at zero or one at its limits, the network can shut off unnecessary latent dimensions, effectively performing a soft band-pass filter. This is particularly useful when the optimal bottleneck size is unknown: one can select a lower latent dimension because the network only uses the dimensions it needs. In testing, this manifests as networks using AERootBlocks needing a latent dimension $d = 32$ for the M4 dataset, but AELG-based networks needing only $d = 16$. For example, a 10-stack TrendWavelet block with AELG root and a latent dimension of 32 using a Daubechies-3 wavelet (`TWAELG_10s_ld32_db3`) uses only 0.48M total parameters while remaining competitive with 15M+ RootBlock models.

2.1.4 Parameter efficiency

The standard width parameters (`g_width`=512, `s_width`=2048, `t_width`=256, `ae_width`=512) govern the main trunk width U for each block family. The `latent_dim` d is an additional hyperparameter that controls only the AE bottleneck and does not change the output dimension $\mathbf{h} \in \mathbb{R}^U$ seen by downstream projection heads.

Table 1: Backbone architecture comparison.

Backbone	Layer topology	Latent operation	Extra params
RootBlock	input $\to U \to U \to U \to U$	None	Baseline
AERootBlock	input $\to U/2 \to d \to U/2 \to U$	Deterministic pass-through	Negligible
AERootBlockLG	Same as AE	$z \cdot \sigma(\gamma)$	d (gate vector)

Table 2: Backbone parameter counts at default widths ($L = 480$).

Backbone	Width U	Latent d	Backbone params	vs. RootBlock
RootBlock (Generic)	512	—	∼810K	1.0×
AERootBlock (Generic)	512	32	∼163K	5.0× fewer
AERootBlock (Generic)	512	16	∼147K	5.5× fewer
AERootBlockLG (Generic)	512	16	∼147K + 16	≈5.5× fewer
RootBlock (Trend)	256	—	∼319K	1.0× (Trend)
AERootBlock (Trend)	256	16	∼46K	6.9× fewer
AERootBlockLG (Trend)	256	16	∼46K + 16	≈6.9× fewer

The backbone parameter count formula for each family is:

- **RootBlock**: $UL + 3U^2$ — four uniform-width layers, no bottleneck.
- **AERootBlock**: $UL/2 + Ud + U^2/2$ — parameters scale roughly as $O(U \cdot d)$ at the bottleneck, which is minor compared to the main width cost.
- **AERootBlockLG**: same as AERootBlock plus d gate parameters — negligible overhead; practical benefit is that d can be halved relative to AERootBlock without accuracy loss.

Because `latent_dim` is typically 5–32, AE variants achieve comparable forecast accuracy with dramatically fewer total backbone parameters than their RootBlock counterparts when `latent_dim` is small and width is constrained. Table 2 gives concrete counts at default widths and a representative backcast length of $L = 480$.

The compression ratio grows with width: wider trunks benefit more from the hourglass bottleneck because the three large $U \times U$ layers in RootBlock scale quadratically while the AE bottleneck scales linearly in d. At Generic/Wavelet width ($U = 512$, $d = 16$) the ratio reaches 5.5×; at Trend width ($U = 256$, $d = 16$) it reaches 6.9×. This makes AE backbones attractive for TrendWavelet and TrendWaveletGeneric configurations, where the Trend trunk width is narrow by design and every saved parameter can be reinvested in stack depth or wavelet family selection, which improves expressivity of the model.

2.2 Branch-specific encoder-decoder designs

The backbone variants described above produce a single shared hidden representation $\mathbf{h}$ that is then split by separate linear heads into backcast and forecast coefficient vectors. A complementary design gives each output branch its own dedicated encoder-decoder sub-network. The `AutoEncoderAE` and `GenericAEBackcastAE` blocks follow this pattern: after the shared AE trunk compresses the input into a trunk latent code z, two independent hourglass sub-branches — one for the backcast path and one for the forecast path — each re-encode z through their own bottleneck before projecting to the output length.

$$z \xrightarrow{\text{enc}_b} z_b \xrightarrow{\text{dec}_b} \hat{\mathbf{x}}, \qquad z \xrightarrow{\text{enc}_f} z_f \xrightarrow{\text{dec}_f} \hat{\mathbf{y}}$$

This gives the network two independent compression points per block. The trunk bottleneck enforces a shared representation of the input window; the branch bottlenecks then clean up each signal independently. The two objectives are separated downstream again to impart more diversity into the learned parameters.

The `latent_dim` hyperparameter controls the branch-local bottleneck size in these blocks, while `thetas_dim` remains reserved for explicit coefficient projections such as the `GenericAEBackcast*`

Table 3: Single-trunk vs. branch-specific AE designs.

Block	Shared trunk	Backcast branch	Forecast branch	`thetas_dim` role
`GenericAE`	AERootBlock	shared $\mathbf{h} \to$ linear	shared $\mathbf{h} \to$ linear	unused
`AutoEncoderAE`	AERootBlock	$z \to$ enc-dec $\to$ output	$z \to$ enc-dec $\to$ output	unused
`GenericAEBackcastAE`	AERootBlock	$z \to$ enc-dec $\to$ output	$z \to$ enc-dec $\to \boldsymbol{\theta}^f \to$ basis	coefficient projection

forecast head. `AutoEncoderAE` omits `thetas_dim` entirely and lets both branch decoders project directly to the target lengths.

The branch-specific design adds approximately $4 \cdot U_b \cdot d_b$ parameters per branch, where U_b is the branch width and d_b the branch latent size. For typical settings ($U_b = U/2$, $d_b = d$), this doubles the AE parameter budget relative to a shared-head block, but keeps the total well below the RootBlock baseline since both the trunk and the branches remain in the hourglass regime.

2.3 Wavelet Family Blocks

The legacy N-BEATS basis-expansion blocks predict coefficient vectors and expand them through a frozen basis into backcast and forecast outputs matching the target lengths. The original interpretable version uses polynomial trend and Fourier seasonality bases. The wavelet blocks replace the Fourier seasonality basis with a fixed, orthonormal discrete wavelet basis. Each block first maps the input window through a RootBlock variant, producing a hidden representation $\mathbf{h}$. Separate trainable linear heads then produce backcast and forecast coefficient vectors,

$$\boldsymbol{\theta}^b = \mathbf{W}_b \mathbf{h}, \qquad \boldsymbol{\theta}^f = \mathbf{W}_f \mathbf{h}$$

For each target length $T \in \{L, H\}$, a wavelet synthesis matrix is constructed at model initialization by applying inverse DWT reconstruction to unit impulses in each wavelet coefficient band. This raw synthesis matrix is then orthogonalized via SVD, and the rows of $V^\top$ form the fixed orthonormal bases $\boldsymbol{\Psi}_b \in \mathbb{R}^{k \times L}$ and $\boldsymbol{\Psi}_f \in \mathbb{R}^{k \times H}$ (one per path). The basis expansion generators therefore evaluate to

$$g_b(\boldsymbol{\theta}^b) = \boldsymbol{\theta}^b \, \boldsymbol{\Psi}_b = \hat{\mathbf{x}}, \qquad g_f(\boldsymbol{\theta}^f) = \boldsymbol{\theta}^f \, \boldsymbol{\Psi}_f = \hat{\mathbf{y}}$$

Because $\boldsymbol{\Psi}_b$ and $\boldsymbol{\Psi}_f$ are frozen at initialization and stored as non-trainable buffers, inference requires only a single matrix multiply per path, which GPUs excel at. A full DWT is never computed at training or inference time. Implementation details are given in Appendix B.

2.3.1 Composition blocks: TrendWavelet and TrendWaveletGeneric

The strongest single-block family in our benchmarks combines a frozen polynomial trend basis with a frozen orthonormal wavelet basis in a single additive decomposition. The `TrendWavelet` block produces the coefficient vector $\boldsymbol{\theta} \in \mathbb{R}^{p+k}$ from the shared hidden representation $\mathbf{h}$, then partitions and expands it:

$$\hat{\mathbf{y}} = \underbrace{\boldsymbol{\theta}^f_{1:p} \, \mathbf{T}_f}_{\text{trend}} + \underbrace{\boldsymbol{\theta}^f_{p+1:p+k} \, \boldsymbol{\Psi}_f}_{\text{wavelet}}$$

where $\mathbf{T}_f \in \mathbb{R}^{p \times H}$ and $\mathbf{T}_b \in \mathbb{R}^{p \times L}$ are the Vandermonde polynomial bases of order p (`trend_thetas_dim`), $\boldsymbol{\Psi}_f \in \mathbb{R}^{k \times H}$ and $\boldsymbol{\Psi}_b \in \mathbb{R}^{k \times L}$ are the orthonormal wavelet bases of rank k (`basis_dim`), and the same additive structure holds for the backcast path. The two bases are complementary by construction: the polynomial captures smooth low-frequency trend while the wavelet captures localized oscillatory residuals.

`TrendWaveletGeneric` extends this to a three-way decomposition by appending a third learned (data-driven) branch of rank r (`generic_dim`):

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^f_{1:p} \, \mathbf{T}_f + \boldsymbol{\theta}^f_{p+1:p+k} \, \boldsymbol{\Psi}_f + \boldsymbol{\theta}^f_{p+k+1:p+k+r} \, \mathbf{G}_f$$

where $\mathbf{G}_f \in \mathbb{R}^{r \times H}$ is a trainable rank-r basis matrix. The two structured bases remain frozen; only coefficients and the generic basis are learned. This gives the block a fall-back data-driven capacity for signal components that neither polynomial nor wavelet bases can represent compactly.

Both block families are fully composable with both AE backbone variants (AERootBlock, AERootBlockLG), yielding the `TrendWaveletAE`, `TrendWaveletAELG`, `TrendWaveletGenericAE`, and `TrendWaveletGenericAELG` configurations benchmarked in Section 4.

2.3.2 Asymmetric and tiered wavelet bases

Two additional hyperparameters control how the wavelet basis is oriented relative to the backcast and forecast windows.

Asymmetric wavelet families. When backcast and forecast lengths differ substantially — as in Traffic-96 ($L = 480$, $H = 96$, ratio $5\times$) — the optimal wavelet support length differs between paths. The `backcast_wavelet_type` and `forecast_wavelet_type` parameters allow independent wavelet family selection per path. A long-support wavelet (e.g. Symlet-20) captures broad low-frequency structure in the long backcast; a short-support wavelet (e.g. Daubechies-3, Coiflet-2) provides compact localized bases for the short forecast. When these overrides are omitted, both paths share the same `wavelet_type`.

Tiered basis offset. The `basis_offset` parameter shifts the starting row of the basis matrix selected from $V^\top$, cycling through the SVD spectrum in successive stacks. In a depth-S model, stack ℓ uses rows $[\ell \cdot \Delta, \ell \cdot \Delta + k)$ of the orthonormal basis, where Δ is a stride derived from `basis_offset`. This spreads each stack's basis coverage across different frequency bands, encouraging the doubly residual stack to decompose the signal rather than repeatedly fitting the same spectral region. Tiered offsets are most effective on the Daily M4 period, where they appear in 8 of the top-10 configurations (Appendix D).

3 Experimental Setup

3.1 Datasets

M4 [**?**] is the primary benchmark (six frequencies, 100,000 series, $H \in \{6, 8, 24, 13, 14, 48\}$). We complement M4 with four out-of-distribution datasets: Tourism (1,311 series, Y/Q/M), PeMS Traffic (862 hourly sensors), Weather (21 channels at 10-min resolution), and the univariate Milk production series.

Table 4: Datasets and evaluation parameters. The paper-sample protocol enforces the in-sample constraint $L_h \cdot H$ [**?**, Appendix D] on the trailing window of each series, with $L_h = 1.5$ for Yearly/Quarterly and $L_h = 10$ otherwise.

Dataset	Frequency	Series	Horizon H	Lookback L	Primary metric
M4 (Y/Q/M/W/D/H)	mixed	100,000	$\{6, 8, 24, 13, 14, 48\}$	$L = 5H$	sMAPE, OWA
Tourism (Y/Q/M)	mixed	1,311	$\{4, 8, 24\}$	$L = 5H$	sMAPE
Traffic-96	hourly	862	96	$5H$	MSE, MAE
Weather-96	10-min	21 ch.	96	$5H$	MSE, MAE
Milk	monthly	1	6	$5H$	sMAPE

3.2 Training Protocols

Every architecture is trained under three complementary protocols (sliding-window with cosine LR; the iteration-based paper-sampling protocol of **?** with MultiStepLR; and the same paper-sampling protocol with `ReduceLROnPlateau` swapped in) over 10 random seeds per (configuration, period, protocol) cell, with pairwise comparisons by Wilcoxon signed-rank test under Bonferroni correction. Full settings and verbatim YAML are in Appendix A.

3.3 Architecture Sweep and Cross-Architecture Transfer

The central hypothesis is that orthonormal wavelet bases and autoencoder bottlenecks enable 60–95% parameter compression relative to paper-faithful Generic stacks while matching or exceeding their

Table 5: Protocol-local winners used for the main comparison. M4 rows use the paper-sample protocol; Weather-96 uses the comprehensive-sweep protocol. Tourism and Milk are summarized in Appendix F. Scores are mean $\pm$ std over the surviving seeds. Asterisks mark `active_g=forecast`, which is not part of the original N-BEATS paper [**?**].

Task	Metric	Best config	Score	Params	n	N-BEATS ref.
M4-Yearly	sMAPE	`T+Coif2V3 30s bdeq`	13.542 ± 0.148	15.2M	10	G-10: 13.564
M4-Quarterly	sMAPE	`NBEATS-IG 10s ag0`	10.312 ± 0.055	19.6M	10	same
M4-Monthly	sMAPE	`TW 30s td3 sym10`	13.240 ± 0.334	6.8M	9	IG-30: 13.307
M4-Weekly	sMAPE	`T+Coif2V3 30s bdeq`	6.735 ± 0.203	15.8M	10	IG-10: 6.974
M4-Daily	sMAPE	`T+Sym10V3 10s tiered`	3.014 ± 0.035	5.3M	10	IG-30: 3.062
M4-Hourly	sMAPE	`NBEATS-IG 30s agf`*	8.758 ± 0.099	43.6M	10	IG-30: 8.906
Weather-96	MSE	`TAE+DB3V3AE 30s ld8`	0.138 ± 0.027	7.1M	10	IG-10: 0.183

forecasting accuracy. We evaluate this under a single 112-configuration *Comprehensive Sweep* (sliding-window protocol) plus a curated 53-configuration subset re-run under both paper-sample variants. The five sweep groups (paper baselines, homogeneous `TrendWavelet` stacks, alternating `T+W` dual-stack patterns, efficiency bottlenecks, and Generic-AE controls) are enumerated in Appendix C; verbatim YAMLs and run scripts ship with the submission. We additionally sweep `active_g` $\in \{\text{False}, \text{forecast}\}$ on the Generic and GenericAELG families to isolate that `lightningnbeats`-extension knob from the wavelet and AE structure.

To confirm the proposed blocks are not specific to the N-BEATS doubly-residual stack, we re-run curated subsets on two other backbones. NHiTS [**?**] adds multi-rate input pooling and hierarchical forecast interpolation around the same block interface; we sweep on Weather-96 and Traffic at $H \in \{96, 192, 336, 720\}$ (MSE loss, 100 epochs, 8 seeds). For Transformer transfer (Section 4.5) we replace each Llama attention projection with a frozen `TrendWavelet` basis on the input-feature axis and the SwiGLU MLP with the same hourglass autoencoder used in our `AERootBlock`, then pretrain a SmolLM2-135M-class model [**?**] from scratch on FineWeb-Edu [**?**] under a 3.0 B-token budget per variant. When block-level effects survive the backbone change, the structured priors are driving them rather than the residual scheduling.

4 Results

All tables apply the strict health filter used in the analysis reports: a run is excluded if `diverged=True`, if sMAPE exceeds 100, or if `best_epoch=0` with sMAPE above 50. Means and standard deviations are computed over surviving seeds. For M4, the main comparison uses the paper-sample protocol to stay close to the N-BEATS training setup of **?**; sliding-window M4 results, plateau-LR variants, and full per-period tables are deferred to Appendix E because absolute sMAPE is not comparable across sampling protocols.

4.1 Per-Period Leaderboards

Table 5 gives the compact leaderboard used for the main claims. Wavelet and TrendWavelet variants win or tie most short- and medium-horizon tasks, but the result is not universal: N-BEATS-I+G [**?**] remains the M4-Quarterly and M4-Hourly reference. M4-Hourly also illustrates an important convention: the winning `agf` configuration uses `active_g=forecast`, a `lightningnbeats` extension rather than the paper-faithful N-BEATS block.

The strongest cross-dataset statement is therefore conditional rather than absolute: structured wavelet bases are a better use of parameters on most tasks in the sweep, but the winning architecture changes with horizon, sampling protocol, and dataset size. Tourism and Milk show the same pattern at smaller scale and are reported in Appendix F.

4.2 Parameter-Efficiency Frontier

Parameter efficiency is the most stable positive result and is the direct test of the central hypothesis stated in Section 3. Table 6 isolates the best sub-1M M4 model for each paper-sample period. These models are not always winners — especially on Weekly — but they repeatedly sit close to the top of the leaderboard at one-tenth to one-fiftieth the parameter count. The clearest example is M4-Hourly:

Table 6: Best sub-1M M4 configurations under the paper-sample protocol. "Gap" is relative to the period winner in Table 5; "Param ratio" compares the period winner's parameter count with the compact model.

Period	Compact config	Params	sMAPE	Gap	Param ratio
Yearly	`TWAE 10s ld32 ag0`	0.48M	13.546 ± 0.102	+0.03%	32×
Quarterly	`TWAE 10s ld32 ag0`	0.49M	10.404 ± 0.028	+0.89%	40×
Monthly	`TWAE 10s ld32 sym10`	0.58M	13.513 ± 0.378	+2.06%	12×
Weekly	`TWAELG 10s ld32 db3`	0.54M	7.252 ± 0.263	+7.68%	29×
Daily	`TWGAELG 10s ld16 db3`	0.52M	3.051 ± 0.049	+1.21%	10×
Hourly	`TWAELG 10s ld32 db3 agf`*	0.85M	8.924 ± 0.129	+1.89%	51×

`TWAELG_10s_ld32_db3_agf` uses 0.85M parameters and trails the 43.6M `NBEATS-IG_30s_agf` reference by only 0.166 sMAPE.

4.3 Stability and Ablations

The negative results are as important as the winners. Overparameterized Generic stacks are unstable: `NBEATS-G_30s_ag0` (paper baseline group of Section 3) has bimodal paper-sample behavior on M4-Quarterly (15.40 ± 9.97), M4-Weekly (8.78 ± 4.61), and M4-Daily (10.21 ± 12.01), even though its best seeds can look competitive. TrendWavelet blocks do not show the same stuck-mode behavior. The small-data version of the same effect shows up on Milk (Appendix F), where AE-family TrendWavelet models cut divergence to the low single digits.

The ablations support five practical rules; the underlying matched-seed Wilcoxon tables are listed in Appendix D.

1. `active_g=forecast` is not a global default. It is decisive on M4-Hourly (9/9 matched pairs improve, 5 significant at $p < 0.05$) and rescues some Generic convergence failures, but is worse or neutral elsewhere and harmful for Weather unified stacks.
2. AE and AELG are statistically similar on M4 at matched configurations; AELG is preferred only when its smaller latent dimension is needed for parameter budget.
3. Tiered basis offsets are strongest on paper-sample M4-Daily but regress on Weekly, so they belong to a Daily-specific recipe rather than the main default.
4. Wavelet family matters across datasets, but the M4 shortlist reduces to `haar`, `db3`, `coif2`, and `sym10`; `coif3` has no per-period M4 SOTA and is dropped from the main defaults.
5. Backbone preference reverses by data regime: RootBlock variants are strongest on M4, while AE variants lead on Weather. Tourism and Milk reproduce this pattern at smaller scale (Appendix F).

4.4 Cross-Architecture Transferability: NHiTS

The block interface transfers more cleanly than the hyperparameters. On the NHiTS backbone [**?**] TrendWavelet and BottleneckGenericAE/AELG families dominate the short-epoch Weather sweep when substituted for the vanilla NHiTS Generic block, with the pooling and hierarchical interpolation schedules unchanged. Short-horizon NHiTS ($H = 96$) favors unified `TrendWavelet`, while long horizons ($H = 336, 720$) favor `BottleneckGenericAE`; `active_g=forecast`, decisive on M4-Hourly, is neutral-to-negative here, and `TrendWaveletGenericVAE` is unexpectedly competitive but we treat that as NHiTS-specific. The cross-architecture takeaway is narrow but genuine: the structured wavelet and AE/AELG components remain useful when lifted out of the N-BEATS doubly-residual schedule, while optimal stack pattern, latent dimension, and scheduler stay dataset- and architecture-dependent. Matching Traffic horizons under the same harness are in Appendix D.

4.5 Cross-Architecture Transferability: Transformer LLMs

The same blocks port into a Transformer with no change in shape. We replace each Llama attention projection with a frozen `TrendWavelet` basis on the input-feature axis, and the SwiGLU MLP with the same hourglass autoencoder used in our `AERootBlock`. Pretraining a SmolLM2-135M-class model [**?**] from scratch on FineWeb-Edu [**?**] under a 3.0 B-token budget, the wavelet variant

`tw_root_fc_db3_64_tiered_silu` (67.2M params, 50.0% compression) reaches a calibrated 24.4 val PPL, Pareto-dominating the pure AE-MLP baseline (69.3M, 26.45) against the 134.5M / 19.82 vanilla reference. A frozen basis on its own is not enough: pure `TrendWavelet` on attention is Pareto-dominated by AE-MLP despite retaining 17% more parameters, and a pure-basis MLP with no learned reduction collapses outright at 250 M tokens. Adding a learned $\mathrm{SiLU}(\mathbf{W}\mathbf{x})$ stage in front of the basis restores competitiveness, the same lesson Section 4.3 draws on M4: a capacity reduction needs a structural prior the optimizer can actually reach. With a pretrained teacher available, the same AE-LG bottleneck at roughly 90% MLP-block compression on Llama-3.2-1B reaches 18.81 PPL after AE pretraining and KD fine-tuning, beating the unmodified 19.45 PPL original. As on M4, where the savings make sub-1M-parameter winners like `TWAELG_10s_ld32_db3` possible (Table 6), in the Transformer the freed budget can be redeposited as deeper stacks, richer composite bases, or wider wavelet sets; full architecture port, trajectories, and distillation pipeline are in Appendix G.

A Paper-Sample Protocol Details

This appendix documents the two paper-sample training protocols in full so that the M4 numbers in Section 4 can be reproduced from a single YAML file. Both variants share the sampling strategy and step budget of the original N-BEATS training procedure; they differ only in the learning-rate scheduler.

A.1 Faithful replication (MultiStepLR)

The faithful variant (`comprehensive_m4_paper_sample.yaml`) is intended as a direct copy of the training procedure described in **?**, Section 5.2 and Appendix D, with no deviations except the small validation-cadence correction described below.

Sampling. Each gradient step draws a batch of 1024 (series, anchor) pairs uniformly at random *with replacement* from the training pool. A series is eligible at step t if it has at least $L + H$ observations; for a chosen series of length T, the anchor index is sampled uniformly from $[T - L_h \cdot H,\ T - H]$, restricting valid windows to the trailing $L_h \cdot H$ time steps. Per-period L_h is fixed to the values in Table 4: $L_h = 1.5$ for Yearly and Quarterly, $L_h = 10$ for Monthly, Weekly, Daily, and Hourly. The lookback length is $L = 5H$.

Optimization. Adam with $\eta_0 = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, no weight decay. sMAPE loss is computed per-step on the 1024-window batch. The gradient budget is 10^5 total steps, with one Lightning epoch defined as 1,000 steps (≈100 paper-epochs).

Learning-rate schedule. MultiStepLR with 10 evenly spaced milestones at $\{10\text{k}, 20\text{k}, \ldots, 100\text{k}\}$ steps, $\gamma = 0.5$. The LR is therefore halved at each 10% boundary of the training budget. This is exactly the schedule used by Oreshkin et al.

Validation and early stopping. Validation is run every 100 training steps (`val_check_interval=100`) on the held-out window of each series. Early stopping monitors validation sMAPE with patience 15 (counted in val-check units, so a minimum of 1,500 steps must elapse before stopping can fire) and $\Delta_{\min} = 10^{-3}$. The sub-epoch validation cadence is the only departure from a strict line-by-line copy of **?**; without it, early stopping fires on the first warmup-corrupted check and collapses to `best_epoch` $\in \{0, 1\}$.

Other settings. `n_blocks_per_stack = 1`, `share_weights = true`, ReLU activations throughout, `forecast_multiplier = 5`, default block widths (`g_width = 512`, `s_width = 2048`, `t_width = 256`).

A.2 Plateau LR variant (ReduceLROnPlateau)

The plateau variant (`comprehensive_m4_paper_sample_plateau.yaml`) is identical to A.1 in every respect except the learning-rate schedule. The motivation is that MultiStepLR commits to a fixed decay schedule independent of how training is actually progressing. A plateau scheduler can

react to validation stalls and find a more aggressive effective decay on cells where the loss surface is well-behaved.

Learning-rate schedule. `ReduceLROnPlateau` with `factor = 0.5`, `patience = 3` (val-check units, so the LR is reduced after 3 consecutive 100-step val checks without improvement of at least $\Delta_{\min} = 10^{-3}$), `cooldown = 1` (one val-check skipped after each reduction before monitoring resumes), `min_lr =` 10^{-6}, `mode = min`, monitored on `val_loss`. The scheduler is stepped on the same cadence as validation (every 100 training steps) so that its trigger logic operates in the same time units as early stopping.

Effect. Empirically the plateau scheduler delivers more LR reductions than MultiStepLR on M4 Quarterly, Monthly, and Daily, and fewer on Yearly and Weekly, which matches the observed differences in the per-period leaderboards (Section 4).

All other settings (sampling, L_h, batch size, step budget, sub-epoch validation, early stopping, optimizer, loss, block widths, weight sharing, seed budget) are inherited unchanged from A.1.

A.3 Why two paper-sample protocols

The N-BEATS paper reports a single training procedure, so any one-protocol comparison conflates the architectural change with whatever artifacts that specific schedule introduces on the new architecture. By running each architecture under both the faithful protocol (Appendix A, MultiStepLR) and the plateau variant (Appendix A, ReduceLROnPlateau) we can attribute observed gains to one of three categories. (i) Gains under both protocols are robust evidence for the architectural claim. (ii) Gains under the faithful protocol only constitute a head-to-head replacement for the published baseline. (iii) Gains under the plateau protocol only are a finding contingent on a more responsive LR schedule, which we report as such rather than as a clean architectural win.

B Hyperparameters and Empirical Defaults

This appendix is the single reference for every hyperparameter exposed by the `lightningnbeats` model classes used in this paper. For each hyperparameter we report (i) the constructor default in the released package, (ii) the value used in the M4 sweeps reported in Section 4, (iii) the swept range when applicable, and (iv) the empirical recommendation that came out of those sweeps. Recommendations are M4-only; out-of-distribution dataset deltas (Tourism, Traffic, Weather, Milk) are summarized in Section 3 and not repeated here. Reproducibility pointers to canonical CSVs and YAML configs are collected in Section B.9.

B.1 Code defaults vs paper-sweep values

Table 7 enumerates every keyword argument of `NBeatsNet.__init__` (`src/lightningnbeats/models.py`, lines 88–313). The constructor defaults define the released package surface; the paper-sweep column lists the value or grid actually used in the experiments of Section 3. Where the paper-sweep value disagrees with the constructor default — most notably `share_weights` — the override is documented at Appendix A (see "Other settings").

Lookback and forecast multiplier. For all M4 cells we use $L = 5H$ (`forecast_multiplier=5`). The paper-sample protocol additionally clamps the valid sampling window to the trailing $L_h \cdot H$ time steps with $L_h = 1.5$ on Yearly/Quarterly and $L_h = 10$ elsewhere; see Appendix A.

B.2 Architectural sweep grids

Table 8 lays out the grid actually executed for each block family in the 112-config Comprehensive Sweep (sliding) and the curated 53-config subset (paper-sample MultiStepLR + plateau). Block-family abbreviations match Section 3: `T+W` = alternating `Trend` + `WaveletV3` (RootBlock), `T+W AELG` = alternating `TrendAELG` + `WaveletV3AELG`, `TW` = unified `TrendWavelet` block, `TWAE/TWAELG` = unified `TrendWaveletAE/TrendWaveletAELG`, `TWGAELG` = `TrendWaveletGenericAELG`.

Table 7: N-BEATS / lightningnbeats hyperparameters: constructor defaults and the values used in this paper's M4 sweeps.

Parameter	Constructor default	Paper sweep value (M4)
Optimization (`_NBeatsBase`)		
`loss`	`SMAPELoss`	`SMAPELoss`
`optimizer_name`	`Adam`	`Adam` (β_1=0.9, β_2=0.999)
`learning_rate`	10^{-3}	10^{-3}
`sum_losses`	`False`	`False`
`kl_weight`	10^{-3}	0.1 for VAE families only
Stack composition (`NBeatsNet`)		
`stack_types`	*required*	per-config (Section 3)
`n_blocks_per_stack`	1	1
`share_weights`	`False`	`True` (paper-sample, Appendix A)
`active_g`	`False`	swept over {`False`, `'forecast'`}
Block widths		
`g_width`	512	512
`s_width`	2048	2048
`t_width`	256	256
`ae_width`	512	512
Block-level priors		
`latent_dim`	5	swept $\{8, 16, 32\}$
`basis_dim`	32	set per-period via `eq_fcast` or $2 \cdot$ `eq_fcast`
`forecast_basis_dim`	`None`	not used on M4
`trend_thetas_dim`	3	swept $\{3, 5\}$
`wavelet_type`	`db3`	swept {haar, db3, coif2, sym10}
`backcast_wavelet_type`	`None`	not used on M4 (Traffic-96 only)
`forecast_wavelet_type`	`None`	not used on M4 (Traffic-96 only)
Cross-stack residuals		
`skip_distance`	0	0
`skip_alpha`	0.1	not exercised on M4

Table 8: Hyperparameter grids per block family. "–" means the parameter does not apply to the family.

Family	n_{stacks}	wavelet	`trend_thetas_dim`	`basis_dim`	`latent_dim`	`active_g`	`share_weights`
`NBEATS-G`	$\{10, 30\}$	–	–	–	–	{F, f}	`True`
`NBEATS-IG`	$\{10, 30\}$	–	3	–	–	{F, f}	`True`
`TW`	$\{10, 30\}$	shortlist	$\{3, 5\}$	{eq, 2·eq}	–	`F`	`True`
`T+W` (RB)	$\{10, 30\}$	shortlist	3	{eq, 2·eq}	–	{F, f}	`True`
`T+W AELG`	$\{10, 30\}$	shortlist	3	`eq`	$\{16, 32\}$	{F, f}	`True`
`TWAE`	10	shortlist	3	`eq`	$\{8, 16, 32\}$	{F, f}	`True`
`TWAELG`	10	shortlist	3	`eq`	$\{16, 32\}$	{F, f}	`True`
`TWGAELG`	10	shortlist	3	`eq`	$\{16, 32\}$	{F, f}	`True`

Wavelet shortlist: {haar, db3, coif2, sym10}. *eq* ≡ *forecast_length.* F ≡ `False`, f ≡ `'forecast'`.

Wavelet basis implementation. For each target length $T \in \{L, H\}$, `_WaveletGeneratorV3` (`src/lightningnbeats/blocks/blocks.py:2621`) builds a raw synthesis matrix of size $T \times T$ by reconstructing each unit impulse in each DWT coefficient band through `pywt.waverec`. The decomposition level is $\min(L_{\max}, 5)$, where $L_{\max} =$ `pywt.dwt_max_level`$(T,$ `dec_len`$)$; when $L_{\max} = 0$ (filter longer than target), we keep level 0 rather than forcing a boundary-corrupted level-1 decomposition. The raw matrix is then SVD-orthogonalized: rows of $V^\top$ form an orthonormal basis ordered low-to-high frequency by singular value. `basis_offset` chooses the starting row and `basis_dim` the row count, so a stack at offset $\ell\,\Delta$ with width k uses $V^\top[\ell\,\Delta : \ell\,\Delta + k,\ :]$. The result is registered as a non-trainable buffer; inference is a single `matmul`.

B.3 Training-protocol hyperparameters (sliding)

The faithful and plateau paper-sample variants are documented in Appendix A. The sliding-window protocol used by the Comprehensive Sweep (`experiments/configs/comprehensive_sweep_m4.yaml`) is: epoch-based training with sMAPE loss, Adam at 10^{-3}, batches drawn by sliding the lookback over each series; `max_epochs`$=$ 75, `min_epochs`$=$ 10, early-stop `patience`$=$ 20 with $\Delta_{\min} = 10^{-3}$; cosine annealing with a 15-epoch linear warmup down to $\eta_{\min} = 10^{-6}$. Sub-epoch validation is not required under sliding because the per-epoch validation pass is on the natural held-out tail of each series.

B.4 Per-period recommendations

Sections B.4.1–B.4.3 are the empirical findings that came out of the sweeps in Section 4. Numbers are taken verbatim from the meta-analysis report `experiments/analysis/analysis_reports/m4_overall_leaderboard_2026-05-03.md`; absolute sMAPE is not comparable across protocols.

B.4.1 LR scheduler by period and protocol

Table 9: Recommended LR scheduler per (period, protocol). Plateau $\equiv$ `ReduceLROnPlateau` with patience $= 3$ val-check units, factor $= 0.5$, cooldown $= 1$, min_lr $= 10^{-6}$. Step-paper $\equiv$ MultiStepLR with 10 evenly spaced milestones, $\gamma = 0.5$. Cosine-warmup $\equiv$ 15-epoch linear warmup then cosine annealing to $\eta_{\min} = 10^{-6}$.

Period	Paper-sample (`nbeats_paper`)	Sliding
Yearly	step-paper (neutral; either works)	cosine-warmup
Quarterly	**plateau** ($\Delta_{\text{step}\to\text{plateau}} = -0.029$, $n = 30$)	cosine-warmup
Monthly	**plateau** ($\Delta = -0.089$, median -0.198, $n = 18$)	cosine-warmup
Weekly	step-paper (plateau regresses by $+0.20$ at $n = 10$, locked out)	cosine-warmup
Daily	**plateau** ($\Delta = -0.028$, $n = 13$)	cosine-warmup
Hourly	step-paper (head-to-head against plateau not yet run on novel families)	cosine-warmup

Mandatory. Plateau LR on `nbeats_paper` requires sub-epoch validation: `val_check_interval=100`, `patience=20` val-check units, $\Delta_{\min} = 10^{-3}$. Without these, early stopping fires on the first warmup-corrupted check and collapses to `best_epoch` $\in \{0, 1\}$ (Appendix A).

B.4.2 Sampling protocol verdict per period

Table 10: Sampling-protocol verdict per M4 period. $\Delta =$ sliding $-$ paper-sample on the best config for each protocol; negative Δ favors sliding. Absolute sMAPE within a row is not comparable across protocols (Appendix A).

Period	H	Best sliding sMAPE	Best paper-sample sMAPE	Δ	Winner
Yearly	6	13.499	13.542	≈ 0	tie
Quarterly	8	10.127	10.313	-0.19	**sliding**
Monthly	18	13.279	13.240	$+0.04$	paper-sample
Weekly	13	6.671	6.735	-0.06	sliding
Daily	14	2.588	3.012	$\mathbf{-0.42}$	**sliding**
Hourly	48	8.587	8.758	-0.17	**sliding**

For paper-faithful comparison to **?** we report paper-sample numbers in the main body. For absolute-sMAPE leaderboards on long-horizon periods (Daily, Hourly, Quarterly), sliding is the stronger protocol. We never mix the two within a leaderboard.

B.4.3 Wavelet family per period

The shortlist {haar, db3, coif2, sym10} is the M4 default. `coif3` is dropped because it produced no per-period SOTA in either protocol despite being tested. Per-period leaders are summarized in Table 11. The cross-period generalist under paper-sample is `sym10` (configuration `T+Sym10V3_10s_tiered_ag0`, mean rank $13.33/108$); under sliding it is `haar` (`T+HaarV3_30s_bd2eq`, mean rank $12.7/112$).

Table 11: Best wavelet family per M4 period within the shortlist. Paper-sample uses the LR scheduler from Table 9; sliding uses cosine-warmup throughout.

Period	Paper-sample best wavelet	Sliding best wavelet
Yearly	db3 / coif2 (within seed noise)	coif2
Quarterly	sym10 (paper baseline NBEATS-IG wins overall)	sym10 / haar
Monthly	sym10 (unified TW)	coif2 (unified TW)
Weekly	coif2 (alternating T+W)	db3 (alternating T+W)
Daily	sym10 (tiered)	paper baseline (NBEATS-G)
Hourly	sym10 (tiered) or paper baseline	paper baseline (NBEATS-IG)

B.5 Block-level recommendations

The following compact rules summarize the empirical findings for each block-level hyperparameter on M4. Each rule is sourced from the meta-analysis report and the sweep CSVs in `experiments/results/m4/`.

`basis_dim`. Use the label `eq_fcast`, i.e. $k = H$, as the default. At convergence (R3, ≥ 50 epochs equivalent) the spread between `eq_fcast`, `lt_bcast`, and `eq_bcast` is < 0.014 sMAPE on M4-Yearly across 14 wavelet families and three datasets; the apparent advantage of larger labels at 10 epochs is a convergence-speed artifact. Avoid `lt_fcast` for $H \leq 8$ (0% R3 survival on M4-Yearly in the AELG sweep). For unified `TrendWavelet` blocks the basis-dim sensitivity is higher (spread 0.37 sMAPE), so `eq_fcast` is even more critical there.

`latent_dim`. For `AERootBlockLG`-backed blocks (`TWAELG`, `TAELG+WV3AELG`, etc.), `latent_dim`$= 16$ is the default ($n \geq 600$ runs, $p < 0.02$ vs 32 on both M4-Yearly and Tourism-Yearly). For plain `AERootBlock`, `latent_dim`$\in \{5, 8\}$ are statistically equivalent (Mann–Whitney $p = 0.71$, $n = 330$ each); `latent_dim`$= 2$ is catastrophically underparameterized ($p < 3 \times 10^{-20}$ vs $\{5, 8\}$). The unified `TrendWaveletAELG` block prefers `latent_dim`$= 16$, with one known catastrophic combination: `db4` + `eq_fcast` + `latent_dim`$= 16$ (sMAPE ~ 76); `db4` at `latent_dim`$= 8$ or with other basis-dim labels is fine.

`trend_thetas_dim`. Use 3 for $H \leq 10$ and 5 for $H \geq 50$. Significant on Tourism-Yearly ($H = 4$, $p = 0.008$, ttd$= 3$) and on Weather-96 ($H = 96$, $p = 0.042$, ttd$= 5$) at convergence. Mid-horizon crossover (Monthly $H = 18$, Weekly $H = 13$, Daily $H = 14$) defaults to 3 on insufficient evidence. The earlier "always ttd$= 3$" finding was a 10-epoch convergence-speed artifact on Weather, reversed at 50 epochs.

Trend backbone hierarchy. On M4 the ordering is RootBlock $>$ AELG $>$ AE for alternating Trend+Wavelet stacks. The plain AE bottleneck homogenizes wavelet-family signal in alternating configurations ($\eta^2 = 0.003$, ns) and is not viable; the learned gate in AELG restores competitiveness. On Weather-96 and Milk this hierarchy reverses (AE $>$ AELG, Mann–Whitney $p = 0.036$ on Weather), but those datasets are out of scope for this appendix.

Block ordering. Always place `Trend*` stacks before `Wavelet*`, `Generic*`, or `Seasonality*` stacks. In the VAE2 study (M4-Yearly + Weather-96) every significantly-different ordering pair — including one Bonferroni-significant pair on M4 (Cohen's $r = 0.83$) — favored Trend-first; no pair favored the reverse. For VAE-family backbones, reversing the order can $2{-}3\times$ the OWA. Treat this as a fixed convention, not a hyperparameter to search.

TrendWavelet vs alternating T+W. Under paper-sample protocol on M4, alternating $T+\langle\text{wav}\rangle$V3 (RootBlock) beats unified `TrendWavelet` by $\approx$ 10 mean-rank points (14.1 vs 24.3 across 53 configs $\times 6$ periods). Default to alternating when $\geq$ 15M parameters is acceptable. Reserve unified `TrendWavelet` (or its AE/AELG variants) for parameter-constrained scenarios: `TWAELG_10s_ld32_db3` achieves 0.48–0.85M parameters and top-5 on Yearly, Daily, and Hourly. Plateau LR has flipped the Monthly winner to unified `TW_30s_td3_bdeq_sym10` (13.240); on every other period alternating is at least co-optimal.

`active_g`**.** The constructor default `False` (paper-faithful, abbreviated `ag0`) is the robust choice on M4. `active_g='forecast'` (`agf`) helps on Hourly only — every paper backbone gains 0.07–0.15 sMAPE there — and ties or loses on every other period. `active_g` is a `lightningnbeats` extension and is not part of the original N-BEATS architecture: configurations marked `ag0` are paper-faithful, configurations marked `agf` apply the extension and are not strictly comparable to **?**.

Other ablations (not exercised on M4). `skip_distance` (cross-stack residual injection) never helps on M4 in the configurations tested and slightly hurts deep stacks; `sum_losses` adds a backcast-reconstruction term ($0.25 \cdot \text{sMAPE}(\hat{x}, 0)$) and is reserved for the convergence study reported in Appendix D. Both default to off in this paper.

B.6 Drop list

Configurations excluded *a priori* from the curated paper-sample subset and from any paper-headline leaderboard. Each is supported by at least one CSV in `experiments/results/m4/`.

- **BottleneckGeneric family** (`BNG*`, `BNAE*`, `BNAELG*`): universally worst on M4.
- **Pure VAE configs** (`GenericVAE_3s_sw0`, etc.): mean sMAPE 55–68 on Q/M/W/D, unusable.
- `NBEATS-G_30s_ag0` **on Q/W/M**: bimodal collapse, sMAPE std 7.4–9.5.
- **All** `*_sd5` **variants** (`skip_distance`$=5$): never help on M4.
- **All** `*_coif3` **variants on M4**: no per-period SOTA across either protocol.
- `_30s_agf` **tiered configs at** `_10s` **depth**: run-to-run divergence outliers.
- **Step-paper LR on Q/M/D**: dominated by plateau LR (Table 9).
- **Pure** `GenericAE_*`**,** `GenericAELG_*`: bottoms out at mean rank ≈ 38.

B.7 Quick-pick decision table

Table 12 is the per-period recipe to copy into a YAML config for a new M4 experiment. Each row reproduces the per-period top cell of the appropriate leaderboard (paper-sample if the column says `nbeats_paper`, sliding otherwise) from `m4_overall_leaderboard_2026-05-03.md`.

Table 12: Best M4 configuration per (period, protocol). All cells are $n = 10$ unless noted. "ttd" = `trend_thetas_dim`; "ld" = `latent_dim`; "bd" = `basis_dim` label; "ag" = `active_g` (0 = `False`, f = `'forecast'`).

Period	Protocol	LR	Block family / config	Depth	Wavelet	ttd	bd	ld	ag	sMAPE
Yearly	paper-sample	plateau	`T+Coif2V3` (RB)	30	coif2	3	`eq`	–	0	13.542
Quarterly	paper-sample	plateau	`NBEATS-IG`	10	–	3	–	–	0	10.313
Monthly	paper-sample	plateau	`TW` (unified)	30	sym10	3	`eq`	–	0	13.240
Weekly	paper-sample	step-paper	`T+Coif2V3` (RB)	30	coif2	3	`eq`	–	0	6.735
Daily	paper-sample	plateau	`T+Sym10V3` (RB, tiered)	10	sym10	3	`eq`	–	0	3.012
Hourly	paper-sample	step-paper	`NBEATS-IG`	30	–	3	–	–	f	8.758
Yearly	sliding	cos-warmup	`TW` (unified)	10	coif2	3	`eq`	–	0	13.499
Quarterly	sliding	cos-warmup	`NBEATS-IG`	10	–	3	–	–	0	10.127
Monthly	sliding	cos-warmup	`TW` (unified)	30	coif2	3	`2eq`	–	0	13.279
Weekly	sliding	cos-warmup	`T+Db3V3` (RB)	30	db3	3	`eq`	–	0	6.671
Daily	sliding	cos-warmup	`NBEATS-G`	30	–	–	–	–	0	2.588
Hourly	sliding	cos-warmup	`NBEATS-IG`	30	–	3	–	–	f	8.587
Sub-1M parameter champions (paper-sample):										
Yearly	paper-sample	plateau	`TWAE_10s_ld32`	10	db3	3	`eq`	32	0	13.546
Daily	paper-sample	plateau	`TWGAELG_10s_ld16_db3`	10	db3	3	`eq`	16	0	3.051

B.8 Cross-period generalists

When a single configuration must run across all six M4 periods (cross-period reporting, ensemble bases, ablation harnesses), three configurations dominate by mean per-period rank:

- **Paper-sample, all-6-period coverage:** `T+Sym10V3_10s_tiered_ag0`, mean rank $13.33/108$, 5.07–6.06M parameters; top-11 on every M4 period.
- **Paper-sample, paper-faithful alternative:** `NBEATS-IG_30s_ag0`, mean rank $16.0/68$, 38–44M parameters. Use when reproducing **?** or when zero divergence is required.
- **Sliding, all-6-period coverage:** `T+HaarV3_30s_bd2eq`, mean rank $12.7/112$, 16.25M parameters; smaller alternative `TALG+HaarV3ALG_30s_ag0` (3.66M, mean rank 21.0).

B.9 Reproducibility

Canonical CSVs (`experiments/results/m4/`). Paper-sample (step-paper LR): `comprehensive_m4_paper_sample_results.csv`, `comprehensive_m4_paper_sample_sym10_fills_results.csv`, `tiered_offset_m4_allperiods_paperlr_results.csv`, `m4_hourly_sym10_tiered_offset_paperlr_results.csv` Paper-sample (plateau LR): `comprehensive_m4_paper_sample_plateau_results.csv`, `tiered_offset_m4_allperiods_results.csv`, `m4_hourly_sym10_tiered_offset_results.csv`. Sliding: `comprehensive_sweep_m4_results.csv`.

Canonical analysis. Meta-leaderboard at `experiments/analysis/analysis_reports/m4_overall_leaderboard_202` regenerable from `experiments/analysis/scripts/m4_overall_leaderboard.py`. The strict health filter `diverged` $\vee$ `smape`$>$`100` $\vee$ (`best_epoch`$=$`0` $\wedge$ `smape`$>$`50`) $\vee$ `smape NaN` is applied to every cell.

Canonical YAML configs (`experiments/configs/`). Paper-sample faithful: `comprehensive_m4_paper_sample.yaml`. Paper-sample plateau: `comprehensive_m4_paper_sample_plateau.yaml`. Sliding: `comprehensive_sweep_m4.yaml`. Tiered: `tiered_offset_m4_allperiods.yaml` (and `m4_hourly_sym10_tiered_offset.yaml` for the dedicated Hourly file).

B.10 Compute resources

All experiments reported in this paper were run on a single workstation: $2\times$ NVIDIA RTX 5090 (32 GB GDDR7), AMD Ryzen Threadripper 7970X (32C/64T), 128 GB RAM, CUDA 13.0, driver 580.126.09. Each individual run uses one GPU; the two GPUs were used to run independent jobs in parallel from a shared work queue. The M4 paper-sample step budget is fixed at 10^5 gradient steps with batch size 1024 (Appendix A); sliding-window M4 runs at most 75 epochs with patience-20 early stopping (Appendix B.3). The Comprehensive Sweep (112 configs $\times$ 6 M4 periods $\times$ 10 seeds) plus the curated paper-sample (53 configs $\times$ 6 $\times$ 10) and plateau (53 $\times$ 6 $\times$ 10) reruns dominate the budget, with Tourism, Traffic, Weather, and Milk sweeps adding a smaller fraction. The from-scratch SmolLM2-135M-class LLM transfer runs (Appendix G) and the Llama-3.2-1B distillation recovery were the most expensive single jobs but a small fraction of total GPU-hours; only one variant per LLM cell was run, and the SmolLM2 wavelet runs were stopped before the full 3.0 B-token budget as noted in Section 4.5.

C Configuration list

This appendix enumerates the YAML configs that drive every sweep cited in the paper. All files live under `experiments/configs/`; verbatim YAML and run scripts ship with the submission.

C.1 M4 sweeps

Sweep groups. The 112-config Comprehensive Sweep (sliding) and 53-config curated paper-sample subset partition into five families: (i) paper baselines (`NBEATS-G`, `NBEATS-IG` at $n_{\text{stacks}} \in \{10, 30\}$ and `active_g` $\in \{\text{F}, \text{f}\}$); (ii) homogeneous `TrendWavelet` stacks

Table 13: M4 sweep configurations. Sliding ≡ `nbeats_sliding` sampling with cosine-warmup LR; paper-sample ≡ `nbeats_paper` iteration-based protocol of Appendix A with either MultiStepLR (faithful) or ReduceLROnPlateau.

Config file	Protocol	Cell count
`comprehensive_sweep_m4.yaml`	sliding (cosine-warmup)	112 configs × 6 periods × 10 seeds
`comprehensive_m4_paper_sample.yaml`	paper-sample (step)	53 configs × 6 periods × 10 seeds
`comprehensive_m4_paper_sample_plateau.yaml`	paper-sample (plateau)	53 configs × 6 periods × 10 seeds
`comprehensive_m4_paper_sample_sym10_fills.yaml`	paper-sample (step)	sym10 fill-ins for the 53-cell subset
`tiered_offset_m4_allperiods.yaml`	paper-sample (plateau)	16 tiered configs × 5 periods (Y/Q/M/W/D) × 1
`tiered_offset_m4_allperiods_paperlr.yaml`	paper-sample (step)	matching MultiStepLR variant
`m4_hourly_sym10_tiered_offset.yaml`	paper-sample (plateau)	8 Hourly tiered configs × 10 seeds
`m4_hourly_sym10_tiered_offset_paperlr.yaml`	paper-sample (step)	Hourly MultiStepLR variant
`nbeats_g.yaml`, `nbeats_ig.yaml`	paper-sample (step)	paper baselines (10s, 30s; `ag0`, `agf`)

(`TW_{10,30}s_td3_{eq,2eq}_{haar,db3,coif2,sym10}`); (iii) alternating `T+W` dual-stack patterns over the same wavelet shortlist (RootBlock, AE, AELG backbones); (iv) efficiency bottlenecks (`TWAE`, `TWAELG`, `TWGAELG` at `latent_dim` $\in \{8, 16, 32\}$, $n_{\text{stacks}} = 10$); (v) Generic-AE controls (`GenericAE`, `GenericAELG`, `BNGenericAE` families). Hyperparameter grids are tabulated in Appendix B.2.

C.2 Out-of-distribution sweeps

Table 14: Out-of-distribution sweep configurations.

Config file	Dataset	Notes
`comprehensive_sweep_tourism.yaml`	Tourism (Y/Q/M)	sliding, sMAPE loss
`omnibus_benchmark_tourism.yaml`	Tourism	cross-family omnibus
`tourism_aelg_sota_confirmation.yaml`	Tourism-Yearly	targeted successive-halving on AELG
`comprehensive_sweep_traffic.yaml`	Traffic-96	sliding, MSE loss, 100 epochs
`traffic_asymmetric_wavelet_study.yaml`	Traffic-96	asymmetric wavelet families ($L = 480$ vs. $H =$
`comprehensive_sweep_weather.yaml`	Weather-96	sliding, MSE loss
`comprehensive_sweep_weather_fm7.yaml`	Weather-96	forecast-multiplier sweep
`omnibus_benchmark_weather.yaml`	Weather	cross-family omnibus
`milk_convergence.yaml`, `milk_convergence_10stack.yaml`	Milk	convergence-rate study (100 seeds)
`unified_benchmark_milk.yaml`	Milk	block-family benchmark
`nhits_benchmark_weather.yaml`	Weather (NHiTS)	NHiTS transferability
`nhits_novel_ae_weather.yaml`	Weather (NHiTS)	novel-block port to NHiTS
`nhits_benchmark_weather_sliding.yaml`	Weather (NHiTS)	sliding variant

C.3 Targeted ablation configs

- `gate_fn_study_m4.yaml`, `gate_fn_study_weather.yaml` — gating-function ablation on AELG.
- `generic_dim_sweep.yaml` — `generic_dim` (rank of learned generic branch in `TrendWaveletGeneric`) ablation.
- `kl_weight_sweep_m4.yaml`, `kl_weight_sweep_weather.yaml` — VAE KL-weight ablation.
- `lg_vae_study_m4.yaml`, `lg_vae_study_large.yaml` — learned-gate vs. VAE comparison.
- `resnet_skip_study{,_v2,_v3}.yaml` — `skip_distance` cross-stack residual injection.
- `ae_trend_study.yaml`, `trendae_study.yaml`, `trendae_ae_alternating.yaml` — Trend backbone hierarchy.
- `trend_seasonality_wavelet_comparison.yaml` — decomposition ablation.
- `tiered_offset_m4_weekly_plateau_validation.yaml` — Weekly plateau-LR tiered re-validation ($n = 10$).

D Additional ablations

This appendix lists the matched-seed Wilcoxon evidence for every empirical rule cited in Section 4.3 and Appendix B.5. All numbers are taken verbatim from the analysis reports under `experiments/analysis/analysis_reports/` and the strict health filter (Section 4) is applied throughout.

D.1 `active_g`: Definition, Instability, and Per-Period Effect

Definition. Standard N-BEATS Generic blocks produce their backcast and forecast estimates via a pair of unconstrained linear projections from the shared trunk hidden state:

$$\hat{\mathbf{b}} = \mathbf{W}_b \mathbf{h}, \qquad \hat{\mathbf{f}} = \mathbf{W}_f \mathbf{h},$$

where $\mathbf{h} \in \mathbb{R}^U$ is the output of the four-layer fully-connected trunk and $\mathbf{W}_b, \mathbf{W}_f$ are learned projection matrices. The `active_g` parameter applies a nonlinear post-activation to these output projections before they are used as basis-expansion coefficients:

$$\hat{\mathbf{b}} = \sigma(\mathbf{W}_b \mathbf{h}), \qquad \hat{\mathbf{f}} = \sigma(\mathbf{W}_f \mathbf{h}),$$

where σ is the model's configured activation function (ReLU by default). The variant `active_g='forecast'` applies the activation only to the forecast projection $\mathbf{W}_f$, leaving the backcast head linear; `active_g=True` activates both heads. The parameter defaults to `False`, preserving the unconstrained linear output of the original N-BEATS architecture [**?**].

Observed instabilities and convergence failures. During a large-scale convergence study ($n = 200$ seeds per configuration, M4-Yearly), overparameterised plain Generic stacks—in particular `NBEATS-G_30s_ag0`—exhibited a pronounced bimodal failure mode: a substantial fraction of random seeds converged to a degenerate fixed point with sMAPE an order of magnitude above the well-converged subset. The same pathology appeared at extreme rates on the small-data Milk series: 63% of 6-stack seeds and 64% of 10-stack seeds were classified as stuck (sMAPE > 5). A less severe version manifested on M4-Yearly at 4.5%.

Setting `active_g` to either `True` or `'forecast'` eliminated the stuck-mode failure *entirely* in every configuration where it was tested. The 63% failure rate on Milk (6-stack) fell to 0%; the 64% rate on Milk (10-stack) fell to 0%; the 4.5% M4-Yearly rate fell to 0% (Table 15). The improvement in convergence reliability is unambiguous and statistically decisive across all datasets examined.

A working hypothesis: unactivated linear outputs as a source of instability. The observation that a single activation function placed after the Generic block's output projections eliminates convergence failures—without any other architectural change—invites a mechanistic explanation. We offer the following as a working hypothesis, and acknowledge that we do not yet have evidence sufficient for a strong theoretical claim.

In the doubly-residual N-BEATS stack, each block decomposes its input by subtracting its backcast from the residual stream and accumulating its forecast into the output. When the final projection is linear and unconstrained, there is no inherent bound on the magnitude of the backcast or forecast signals produced in early training. If a block produces a large-magnitude backcast by chance at initialisation, it over-subtracts the input residual; the corrupted signal passed to the next block can then be further amplified by that block's own unconstrained projections. In deep stacks this cascade can compound across many layers, driving the model into a region from which a standard Adam optimiser with a cosine or step learning-rate schedule cannot recover.

Applying an activation to the output projection bounds the projected signal—ReLU, for instance, restricts it to the non-negative half-space—and effectively regularises the magnitude of the backcast and forecast estimates throughout early training. We conjecture that this soft amplitude bound prevents the runaway signal cascade responsible for bimodal stuck modes, and that the benefit is greatest precisely where the doubly-residual accumulation is deepest (many stacks) or training data are scarcest (Milk, a single univariate series), so that gradient information is too weak to self-correct the corrupted residual chain.

This hypothesis is broadly consistent with established observations in deep network optimisation: unrestricted linear outputs are known to produce gradient-scale instabilities in certain architectures [**??**],

and activation functions have historically served a stabilising role beyond their role in representational capacity [**?**]. We stress, however, that the causal mechanism here is specific to the doubly-residual structure of N-BEATS; whether linear output instability generalises to other architectures with different signal-routing conventions remains an open question.

Structural regularisation in AE and Wavelet blocks as an implicit substitute. If the instability hypothesis is correct, then any architectural mechanism that inherently bounds the output-projection magnitude should render `active_g` unnecessary—and this is precisely what is observed for AE-family and Wavelet-family blocks.

Autoencoder bottleneck. In `AERootBlock`-backed blocks (`GenericAE`, `TrendAE`, `TrendWaveletAE`, and their LG/VAE descendants), the backcast and forecast are not produced by a direct linear projection from the full-width trunk. Instead, the hidden state is first compressed through a deep bottleneck—hidden $\rightarrow$ hidden$/2$ $\rightarrow$ latent_dim—before being decoded back to the output. Because the latent dimension is typically far smaller than the trunk width (e.g. `latent_dim`$= 16$ against `g_width`$= 512$), the information capacity of the bottleneck is intrinsically limited, and the resulting output projection cannot produce arbitrarily large signals regardless of weight initialisation. The bottleneck compression acts as an implicit amplitude regulariser on the block's output—serving the same stabilising role as `active_g` but through structure rather than nonlinearity. Empirically, AE-family TrendWavelet models on the Milk benchmark show stuck-mode rates in the low single digits even with `active_g=False`, compared with 63–64% for their plain Generic counterparts.

Wavelet basis expansion. In `WaveletV3`-backed blocks, the forecast and backcast are computed as $\hat{\mathbf{f}} = \boldsymbol{\theta}_f \mathbf{B}_f$ and $\hat{\mathbf{b}} = \boldsymbol{\theta}_b \mathbf{B}_b$, where $\mathbf{B}_f$ and $\mathbf{B}_b$ are fixed, SVD-orthonormalised DWT synthesis matrices. Because the rows of $\mathbf{B}$ are orthonormal (each has unit ℓ_2 norm), the energy of the output is controlled directly and exclusively by the coefficient vector $\boldsymbol{\theta}$; the basis itself cannot amplify the signal. Furthermore, $\boldsymbol{\theta}$ is produced by a low-rank projection from the trunk (dimension `basis_dim` $\ll$ hidden), creating a second compression stage analogous to the AE bottleneck. The combination of a rank-limited coefficient projection and an energy-preserving orthonormal synthesis matrix structurally prevents the runaway output-magnitude growth that triggers stuck modes in unconstrained Generic blocks.

Taken together, the convergence data for AE and Wavelet blocks reinforce the hypothesis from the opposite direction: architectures that impose implicit bounds on output magnitude—whether through information bottlenecks or orthonormal bases—do not require `active_g` to achieve reliable convergence. `active_g` is best understood as an explicit, lightweight substitute for the implicit regularisation that AE and Wavelet structure provide natively. Its residual accuracy cost (Table 15) reflects the tighter constraint that a nonlinear activation imposes compared with a soft bottleneck or orthonormal projection, and suggests that structural regularisation is the more principled long-term solution.

Trade-offs. The stability benefit of `active_g` comes at an accuracy cost in most regimes. Even on datasets where it eliminates stuck modes entirely—Milk 6-stack, Milk 10-stack—the mean sMAPE of the converged subset increases, indicating that the activation constrains the model's representational range in ways that are detrimental once optimisation is no longer the binding constraint. On M4, every period except Hourly shows a statistically significant accuracy regression under `agf`. The Hourly gain is genuine and robust ($9/9$ matched pairs improve, 5 at $p < 0.05$), which may reflect properties specific to that period's long horizon and high-frequency structure. Our recommendation is therefore to treat `active_g` as a targeted intervention rather than a global default: use `ag0` for paper-faithful reproducibility; switch to `agf` for M4-Hourly; apply `active_g=True` only when bimodal convergence failures are observed and convergence reliability takes priority over converged-subset accuracy.

Table 15 summarises the quantitative outcome: `active_g` is a Mann–Whitney win on Hourly (where it is decisive) and a Mann–Whitney loss on every other period, and on small-data Milk it eliminates a 63% bimodal failure mode while paying an absolute sMAPE cost on the converged subset.

`active_g=True` **vs.** `active_g='forecast'`**.** The two variants are statistically indistinguishable at $n = 200$ on M4-Yearly ($\Delta = -0.023$, $p = 0.154$) and Tourism-Yearly ($\Delta = +0.066$, $p = 0.087$), so the `'forecast'` variant is preferred only because it isolates the gate to the forecast head.

Table 15: `active_g` effect by dataset, paired Mann–Whitney. "Quality cost" is the absolute sMAPE delta on the converged subset of seeds; the convergence column is the failure-rate reduction. "Converged subset" applies a dataset-specific stuck-mode filter beyond the strict health filter of Section 4 ($\mathrm{sMAPE} < 20$ on M4-Yearly, $\mathrm{sMAPE} < 30$ on Tourism-Yearly, $\mathrm{sMAPE} < 5$ on Milk) so that bimodal stuck seeds do not contaminate the converged-mean estimate.

Dataset	Baseline sMAPE	`active_g` sMAPE	Δ	MWU p	Convergence effect
Milk 6-stack	1.685 (n=37)	2.500 (n=100)	$+0.815$	$< 10^{-6}$	$63\% \to 0\%$ stuck
Milk 10-stack	1.797 (n=36)	2.455 (n=100)	$+0.659$	$< 10^{-5}$	$64\% \to 0\%$ stuck
M4-Yearly	13.616 (n=191)	13.669 (n=200)	$+0.053$	$< 10^{-4}$	$4.5\% \to 0\%$ stuck
Tourism-Yearly	21.539 (n=197)	21.374 (n=200)	-0.165	1.8×10^{-4}	$1.5\% \to 0\%$ stuck

Per-period verdict on M4 paper-sample. Hourly: $9/9$ matched pairs improve under `agf` (5 significant at $p < 0.05$); Hourly in the canonical leaderboard is -0.148 sMAPE (`NBEATS-IG_30s_agf` 8.758 vs. `ag0` 8.906). Yearly/Quarterly/Monthly/Weekly/Daily: `agf` ties or loses on every protocol-LR cell tested. Treat `active_g='forecast'` as a Hourly-only intervention.

D.2 Tiered basis-offset: per-period verdict

The matched-seed (paper-sample, plateau LR) tiered vs. non-tiered comparison from `tiered_offset_m4_allperiods_analysis.md`:

Table 16: Tiered basis-offset, matched against the closest non-tiered baseline. "Better" counts paired wins for the tiered config; significance is paired Wilcoxon at $p < 0.05$.

Period	n_{pairs} tiered $<$ base	Avg. Δ sMAPE	Sig. tiered<base	Sig. base<tiered
Yearly	$7/11$	-0.020	1 (`TAE+DB3V3AE_30s`)	0
Quarterly	$8/11$	-0.020	0	0
Monthly	$7/11$	-0.105	1 (`T+DB3V3_10s_ag0`)	0
Weekly	$2/11$	$\mathbf{+0.130}$	0	0
Daily	$\mathbf{11/11}$	$\mathbf{-0.049}$	**4**	0

The strongest single Daily wins (Cliff's $d \in [0.54, 0.78]$) are `TAELG+Sym10V3AELG_10s_tiered` ($\Delta = -0.080$, $p = 0.0036$, $d = 0.78$), `TAE+DB3V3AE_30s_tiered` ($\Delta = -0.074$, $p = 0.0036$, $d = 0.78$), `TAELG+DB3V3AELG_10s_tiered` ($\Delta = -0.067$, $p = 0.0312$, $d = 0.58$), `TAELG+Sym10V3AELG_30s_tiered` ($\Delta = -0.043$, $p = 0.0452$, $d = 0.54$). Weekly is a real cascade reversal at $H = 13$, confirmed by an $n = 10$ plateau-LR re-validation in `tiered_offset_m4_weekly_plateau_validation_results.csv`.

D.3 LR scheduler: paper-sample matched pairs

Plateau vs. MultiStepLR matched-pair deltas on the 53-config paper-sample subset:

Table 17: Paper-sample plateau-vs-step matched pairs (Δ = plateau $-$ step). Negative Δ favors plateau.

Period	Mean Δ	Median Δ	n_{pairs}
Yearly	$+0.007$	$+0.021$	52
Quarterly	$\mathbf{-0.029}$	-0.032	30
Monthly	$\mathbf{-0.089}$	-0.198	18
Weekly	$+0.089$	$+0.080$	11
Daily	$\mathbf{-0.028}$	-0.028	13

The Hourly scheduler choice is canonicalized at the cell level rather than the period level: `T+Sym10V3_10s_tiered_ag0` (descend) prefers plateau (8.922 vs. 9.090, $p = 0.026$, $n = 10$), while `TWAE_10s_td3_sym10_ld16_tiered` (descend) prefers step_paper (9.243 vs. 9.439, $p = 0.004$, $n = 10$). The leaderboard cell for Hourly therefore mixes scheduler choice by config.

D.4 Traffic-96 NHiTS transferability

Curated subsets of the M4 wavelet/AE/AELG block library were re-run on the NHiTS backbone [**?**] for Traffic at $H \in \{96, 192, 336, 720\}$ under MSE loss, 100 epochs, and 8 seeds. The pooling and hierarchical interpolation schedules are inherited from the published NHiTS recipe. Short-horizon ($H = 96$) winners are unified `TrendWavelet` (sym10 / coif2 wavelet shortlist), long-horizon ($H \geq 336$) winners are `BottleneckGenericAE`/`BottleneckGenericAELG`. Asymmetric wavelet families (Section 2.3, "Asymmetric wavelet families") are most useful at $H = 96$ where the L/H ratio is largest. Per-cell numbers and the matching Weather-96 NHiTS sweep are recorded in `nhits_benchmark_weather_results.csv` and `nhits_benchmark_traffic_results.csv`.

E M4 full per-period tables

This appendix lists the full top-5 leaderboard for every (period, protocol) cell summarized in Table 5. Numbers are taken verbatim from `m4_overall_leaderboard_2026-05-03.md`; the strict health filter of Section 4 is applied. Within each period the paper-sample table mixes step-paper and plateau LR (the per-cell scheduler is shown in the *LR* column); the sliding table uses cosine-warmup throughout.

E.1 M4-Yearly ($H = 6$, $L = 30$)

Table 18: M4-Yearly top-5: paper-sample (left) and sliding (right).

Paper-sample					
#	Config	LR	sMAPE ± std	Params	n
1	`T+Coif2V3_30s_bdeq`	plateau	$\mathbf{13.542 \pm 0.148}$	15.24M	10
2	`TWAE_10s_ld32_ag0`	plateau	13.546 ± 0.102	0.48M	10
3	`TALG+DB3V3ALG_10s_ag0`	step	13.550 ± 0.096	1.04M	10
4	`TWAE_10s_ld32_sym10_ag0`	plateau	13.553 ± 0.165	0.48M	10
5	`T+DB3V3_10s_tiered_agf`	step	13.554 ± 0.101	5.07M	10
Sliding					
1	`TW_10s_td3_bdeq_coif2`	cos-warmup	$\mathbf{13.499 \pm 0.057}$	2.08M	10
2	`TALG+Sym10V3ALG_30s_agf`	cos-warmup	13.504 ± 0.103	3.13M	10
3	`TALG+Db3V3ALG_30s_ag0`	cos-warmup	13.507 ± 0.080	3.13M	10
4	`TWAELG_10s_ld16_coif2_agf`	cos-warmup	13.524 ± 0.060	0.44M	10
5	`T+Db3V3_30s_bd2eq`	cos-warmup	13.529 ± 0.115	15.29M	10

E.2 M4-Quarterly ($H = 8$, $L = 40$)

Table 19: M4-Quarterly top-5: paper-sample (left) and sliding (right).

Paper-sample					
#	Config	LR	sMAPE ± std	Params	n
1	`NBEATS-IG_10s_ag0`	plateau	$\mathbf{10.313 \pm 0.055}$	19.64M	10
2	`T+Sym10V3_10s_tiered_ag0`	plateau	10.325 ± 0.053	5.11M	10
3	`T+DB3V3_30s_tiered_ag0`	step	10.345 ± 0.065	15.34M	10
4	`NBEATS-IG_10s_agf`	plateau	10.354 ± 0.115	19.64M	10
5	`T+DB3V3_10s_tiered_ag0`	plateau	10.356 ± 0.053	5.11M	10
Sliding					
1	`NBEATS-IG_10s_ag0`	cos-warmup	$\mathbf{10.127 \pm 0.068}$	19.64M	10
2	`T+Sym10V3_30s_bd2eq`	cos-warmup	10.127 ± 0.051	15.45M	10
3	`TALG+HaarV3ALG_30s_ag0`	cos-warmup	10.144 ± 0.048	3.28M	10
4	`T+Coif2V3_30s_bd2eq`	cos-warmup	10.146 ± 0.045	15.45M	10
5	`TW_10s_td3_bdeq_coif2`	cos-warmup	10.147 ± 0.056	2.11M	10

Table 20: M4-Monthly top-5: paper-sample (left) and sliding (right).

Paper-sample					
#	Config	LR	sMAPE ± std	Params	n
1	`TW_30s_td3_bdeq_sym10`	plateau	$\mathbf{13.240 \pm 0.334}$	6.78M	9
2	`TW_30s_td3_bdeq_coif3`	plateau	13.296 ± 0.182	6.78M	5
3	`NBEATS-IG_30s_ag0`	plateau	13.307 ± 0.388	38.13M	9
4	`NBEATS-G_10s_ag0`	plateau	13.312 ± 0.252	8.90M	10
5	`NBEATS-G_30s_agf`	plateau	13.337 ± 0.417	26.70M	10
Sliding					
1	`TW_30s_td3_bd2eq_coif2`	cos-warmup	$\mathbf{13.279 \pm 0.303}$	7.08M	10
2	`T+HaarV3_30s_bd2eq`	cos-warmup	13.308 ± 0.242	16.25M	10
3	`NBEATS-IG_10s_ag0`	cos-warmup	13.309 ± 0.255	20.33M	10
4	`TALG+Coif2V3ALG_30s_ag0`	cos-warmup	13.309 ± 0.251	4.03M	10
5	`TW_30s_td3_bdeq_sym10`	cos-warmup	13.314 ± 0.222	6.78M	10

Table 21: M4-Weekly top-5 paper-sample (step_paper LR) and best plateau-LR cell (n=10 re-validation), plus sliding top-5.

Paper-sample (step_paper LR)					
#	Config	LR	sMAPE ± std	Params	n
1	`T+Coif2V3_30s_bdeq`	step	$\mathbf{6.735 \pm 0.203}$	15.75M	10
2	`T+Db3V3_30s_bdeq`	step	6.737 ± 0.256	15.75M	10
3	`T+Sym10V3_30s_bdeq`	step	6.858 ± 0.273	15.75M	10
4	`T+Sym10V3_30s_tiered_ag0`	step	6.907 ± 0.293	15.68M	10
5	`T+Coif2V3_10s_bdeq`	step	6.919 ± 0.424	5.25M	10
Paper-sample (plateau LR re-validation, regression)					
1	`TAE+Sym10V3AE_30s_tiered`	plateau	6.935 ± 0.223	3.37M	10
2	`TAE+DB3V3AE_30s_tiered`	plateau	6.955 ± 0.233	3.37M	10
Sliding					
1	`T+Db3V3_30s_bdeq`	cos-warmup	$\mathbf{6.671 \pm 0.208}$	15.75M	10
2	`TALG+HaarV3ALG_30s_ag0`	cos-warmup	6.673 ± 0.129	3.66M	10
3	`T+HaarV3_30s_bdeq`	cos-warmup	6.675 ± 0.185	15.75M	10
4	`T+HaarV3_30s_bd2eq`	cos-warmup	6.685 ± 0.193	15.85M	10
5	`T+Sym10V3_30s_bdeq`	cos-warmup	6.686 ± 0.168	15.75M	10

E.3 M4-Monthly ($H = 18$, $L = 90$)

E.4 M4-Weekly ($H = 13$, $L = 65$)

E.5 M4-Daily ($H = 14$, $L = 70$)

E.6 M4-Hourly ($H = 48$, $L = 240$)

E.7 Cross-period generalist mean ranks

F Tourism and Milk Out-of-Distribution Results

This appendix collects the Tourism and Milk evidence summarized in Section 4.1 and Section 4.3. Both datasets are small relative to M4 and serve as out-of-distribution checks rather than as primary benchmarks.

F.1 Tourism (Y/Q/M)

Tourism uses a column-oriented data layout (1,311 series, horizons $H \in \{4, 8, 24\}$, $L = 5H$, sMAPE loss) and is evaluated under the same protocol as M4 but with two orders of magnitude fewer series. Table 26 lists the protocol-local winners we observed under the comprehensive-sweep protocol. Tourism-Yearly was also probed by a targeted successive-halving study outside the main 112-config sweep; the best targeted configuration (`TrendWaveletAELG_coif3_eq_bcast_td3_ld16`, sMAPE 20.864) sits 4.2% below the 112-config sweep winner reported here, so the Tourism-Yearly row should not be read as a final SOTA claim.

Table 22: M4-Daily top-5: paper-sample (left) and sliding (right).

Paper-sample					
#	Config	LR	sMAPE $\pm$ std	Params	n
1	`T+Sym10V3_10s_tiered_ag0`	plateau	$\mathbf{3.014 \pm 0.035}$	5.25M	10
2	`T+DB3V3_10s_tiered_ag0`	plateau	3.023 ± 0.039	5.25M	10
3	`T+Sym10V3_30s_tiered_ag0`	plateau	3.031 ± 0.040	15.75M	10
4	`TAELG+Sym10V3AELG_30s_tiered`	plateau	3.035 ± 0.033	3.41M	10
5	`TAELG+Coif2V3ALG_30s_ag0`	step	3.036 ± 0.034	3.73M	10
Sliding					
1	`NBEATS-G_30s_ag0`	cos-warmup	$\mathbf{2.588 \pm 0.081}$	26.02M	10
2	`NBEATS-IG_30s_ag0`	cos-warmup	2.599 ± 0.027	37.40M	10
3	`NBEATS-G_30s_agf`	cos-warmup	2.605 ± 0.052	26.02M	10
4	`T+Db3V3_30s_bd2eq`	cos-warmup	2.709 ± 0.052	15.93M	19
5	`T+Db3V3_30s_bdeq`	cos-warmup	2.711 ± 0.045	15.82M	17

Table 23: M4-Hourly top-10 paper-sample (mixed scheduler choice; canonicalized in Section 4.1) and sliding top-5.

Paper-sample					
#	Config	LR	sMAPE $\pm$ std	Params	n
1	`NBEATS-IG_30s_agf`	step	$\mathbf{8.758 \pm 0.099}$	43.58M	10
2	`NBEATS-G_30s_agf`	step	8.862 ± 0.085	31.76M	10
3	`NBEATS-IG_10s_agf`	step	8.893 ± 0.106	22.40M	10
4	`NBEATS-IG_30s_ag0`	step	8.906 ± 0.083	43.58M	10
5	`T+Sym10V3_10s_tiered_ag0` (desc.)	plateau	8.922 ± 0.113	6.06M	10
6	`TWAELG_10s_ld32_db3_agf`	step	8.924 ± 0.129	0.85M	10
7	`TWAELG_10s_ld32_sym10_agf`	step	8.933 ± 0.110	0.85M	10
8	`NBEATS-G_30s_ag0`	step	8.934 ± 0.110	31.76M	10
9	`TWAELG_10s_ld16_db3_agf`	step	8.946 ± 0.098	0.81M	10
10	`TWAE_10s_ld32_agf`	step	8.953 ± 0.099	0.85M	10
Sliding					
1	`NBEATS-IG_30s_agf`	cos-warmup	$\mathbf{8.587 \pm 0.080}$	43.58M	10
2	`NBEATS-IG_10s_agf`	cos-warmup	8.629 ± 0.076	22.40M	10
3	`TAE+DB3V3AE_30s_ld16_agf`	cos-warmup	8.673 ± 0.082	5.42M	10
4	`NBEATS-IG_30s_ag0`	cos-warmup	8.680 ± 0.104	43.58M	10
5	`TALG+Coif2V3ALG_30s_agf`	cos-warmup	8.690 ± 0.095	5.42M	10

The qualitative picture matches the M4 sweep at smaller scale. RootBlock-backed unified `TrendWavelet` blocks lead at the small horizons ($H = 4, 8$); AE-family blocks become more competitive at the larger Tourism-Monthly horizon ($H = 24$). The targeted Yearly study additionally confirmed two M4 defaults on Tourism: `trend_thetas_dim`$= 3$ at $H = 4$ ($p = 0.008$) and `latent_dim`$= 16$ for AELG-backed blocks ($p < 0.02$), both reported in Appendix B.

F.2 Milk

The univariate Milk production series is the small-data stress test (one series, $H = 6$, $L = 5H$, sMAPE loss). The 10-stack winner is `TALG+DB3V3ALG_10s` at sMAPE 1.512 ± 0.572, 1.0M parameters, 10 surviving seeds, against an N-BEATS-IG-10 reference of 1.785. The Milk sweep also reproduces the small-data version of the Generic-stack stability problem reported in Section 4.3: RootBlock and Generic baselines show high run-to-run divergence, while AE-family TrendWavelet models cut divergence to the low single digits and produce the reliable winner above. The convergence-rate breakdown by configuration is recorded in `experiments/results/milk_convergence_10stack/milk_convergence_10stack_results.csv`.

F.3 Backbone-by-regime summary

Across the four out-of-distribution datasets, RootBlock variants are the most reliable choice on small- and medium-horizon problems with many series (M4, Tourism). AE-family bottlenecks lead on long-horizon multivariate Weather and on the small-data Milk regime. The trend-backbone hierarchy paragraph in Appendix B (§B, "Trend backbone hierarchy") gives the matched-seed Wilcoxon evidence for the M4 ordering and the Weather reversal.

Table 24: Top-5 paper-sample generalists by mean rank across the six M4 periods. Ranks are within the 108-config paper-sample pool. "–" means the config did not run on that period.

#	Config	Y	Q	M	W	D	H	Mean rank
1	`T+Sym10V3_10s_tiered_ag0`	11	6	41	16	1	5	**13.33**
2	`T+DB3V3_10s_tiered_ag0`	31	5	7	29	3	–	15.00
3	`T+DB3V3_30s_tiered_ag0`	15	7	13	28	13	–	15.20
4	`NBEATS-IG_30s_ag0`	30	13	3	22	24	4	16.00
5	`T+Sym10V3_30s_tiered_ag0`	37	17	28	4	6	–	18.40

Table 25: Top-5 sliding generalists by mean rank across the six M4 periods. Ranks are within the 112-config sliding pool.

#	Config	Y	Q	M	W	D	H	Mean rank
1	`T+HaarV3_30s_bd2eq`	31	20	2	4	9	10	**12.7**
2	`NBEATS-IG_30s_ag0`	41	13	20	28	2	4	18.0
3	`NBEATS-IG_10s_ag0`	22	1	3	55	17	12	18.3
3	`T+Db3V3_30s_bd2eq`	5	9	26	51	4	15	18.3
3	`TW_30s_td3_bdeq_db3`	25	24	8	8	12	33	18.3

G TrendWavelet and AE-MLP on a Transformer LLM

This appendix documents the cross-architecture port summarized in Section 4.5. The full empirical record lives in the sister repository `pellm`; the four primary source reports are listed at the end of this appendix.

G.1 Architecture port

We replace two block types in a Llama-architecture Transformer with the time-series blocks introduced in Section 2. The attention projections (`q_proj`, `k_proj`, `v_proj`, `o_proj`) are replaced by a frozen `TrendWavelet` basis acting on the input-feature axis of the projection weight: 4 polynomial trend components and 28 db3 wavelet components per row, with the basis stored as non-trainable buffers exactly as in Section 2.3. The SwiGLU MLP block is replaced by the same hourglass autoencoder used for `AERootBlock` and `AERootBlockLG` ($\text{hidden} \to \text{hidden}/2 \to d \to \text{hidden}/2 \to \text{hidden}$, with $d = 256$ for the SmolLM2-class runs and $d = 512$ for the Llama-3.2-1B distillation run). Embedding, layer-norm, and residual scaffolding are unchanged.

G.2 From-scratch SmolLM2-135M trajectory

We pretrain each variant from scratch on FineWeb-Edu under a fixed 3.0 B-token budget, batch size 1024, mixed-precision AdamW, cosine learning rate schedule with warmup. Only the `baseline` and `ae_mlp` runs reached the full budget; the wavelet variants were stopped at the token counts shown below and the 3.0 B-token endpoint is a calibrated log-linear projection from the late-training slope (calibration anchor: $+0.057$ nats, measured on the two completed runs to remove the cosine-tail flattening bias).

Two negative controls confirm the inductive-prior reading. A frozen wavelet basis with no learned reduction in the MLP (`tw_root_db3_64_silu`) collapses at 250 M tokens (PPL 112) and was aborted. A frozen wavelet basis on attention with no learned reduction (`trendwavelet_db3_32`) keeps 17% more parameters than `ae_mlp` but is Pareto-dominated at every matched-token point. The competitive variant (`tw_root_fc_db3_64_tiered_silu` in Table 28) is the one that adds a learned $\mathrm{SiLU}(\mathbf{W}\mathbf{x})$ stage in front of the basis. Macro-average zero-shot accuracy across seven `lm-evaluation-harness` tasks [**?**] at 3.0 B tokens is 38.51% for `baseline`, 35.43% for `ae_mlp`, and 34.95% for the wavelet variant at its 1.85 B-token checkpoint, with LAMBADA carrying most of the regression; per-task numbers are in the source pellm reports.

G.3 Distillation recovery on Llama-3.2-1B

The same AE-LG bottleneck behaves differently when a teacher is available. We swap a single layer-15 SwiGLU MLP in Llama-3.2-1B [**?**] for a `PEBottleneckMLPLG` block at $d = 512$, then run a two-phase pipeline: Phase 1 caches teacher MLP activations and trains the AE block to reconstruct

Table 26: Tourism protocol-local winners (comprehensive-sweep protocol, mean $\pm$ std over surviving seeds).

Period	Metric	Best config	Score	Params	n	N-BEATS ref.
Tourism-Y	sMAPE	`TW 10s td3 db3`	21.773 ± 0.384	2.0M	10	IG-10*: 22.180

Table 27: Parameter counts for the SmolLM2-135M-class port. The MLP block accounts for 59% of the baseline parameter count and is the primary compression target. Total compression of the variants is roughly proportional to MLP-block compression.

Variant	Total	MLP block	Attention	Embed + LM head
baseline	134.5M	79.6M	26.5M	28.3M
`ae_mlp`	69.3M	14.4M	26.5M	28.3M
`tw_root_fc_db3_64_tiered_silu`	67.2M	12.3M	26.5M	28.3M

them under MSE; Phase 2 fine-tunes end-to-end with logit-level KD against the unmodified Llama (KD weight $\alpha = 1.0$, temperature 2.0, FineWeb 50k samples).

The recovered model carries roughly 90% MLP-block compression on the swapped layer at 18.81 PPL, -3.3% relative to the unmodified original. Read together with the from-scratch SmolLM2 result, this is consistent with the M4 finding that capacity reductions need a structural prior the optimizer can actually reach: from random init the basis is unreachable, while a teacher (lstsq weight projection or KD logits) provides the rotation the basis is good at decoding.

G.4 Reinvesting saved parameters

The parameter saving is not the headline result on its own; the headline result is what the saved parameters can be redeposited into. On M4 the same trick produces sub-1M-parameter winners like `TWAELG_10s_ld32_db3` (0.48M params, top-5 on Yearly, Daily, and Hourly under the paper-sample protocol; Table 6). On NHiTS the same blocks transfer cleanly to the multi-rate pooled backbone (Section 4.4). On the Transformer the 50% block-budget saving creates room for richer composite bases (`tw_root_fc_post_fc` with a learned post-basis mixer projects 24.1 PPL at 43.5% compression, slightly better than the 50%-compression headline variant), tiered per-layer offset schedules, and depth, without exceeding the original parameter budget.

G.5 Caveats and reproducibility

Single seed per variant in the SmolLM2-class runs; cross-seed standard deviation is unmeasured. Recipe drift between the legacy 2-GPU DDP runs (`baseline`, `ae_mlp`) and the 1-GPU runs (wavelet variants) is absorbed by the calibration anchor described above; reported gaps are an order of magnitude larger than any plausible drift. Only `baseline` and `ae_mlp` reached the full 3.0 B-token budget; wavelet rows in Table 28 are partial-checkpoint values, with the 3.0 B endpoint a projection. Source reports with the full projection methodology, calibration arithmetic, per-task lm-eval breakdowns, AE-pretraining init hierarchy, and fc2/fc3 SVD rule are at `pellm/evals/smol_replacement_paper/preliminary_feasibility_report.md`, `analysis_tw_root_fc_vs_ae_mlp_vs_baseline.md`, `analysis_ae_pretraining.md`, and `experiments/analysis/analysis_reports/pellm_validation_and_theory_revision.md`.

H Limitations

The architectural sweep is centered on the M4 benchmark; while we include Weather and Traffic as primary out-of-distribution checks (with Tourism and Milk in Appendix F), the breadth of structured-prior conclusions outside M4 is bounded by the size of those auxiliary sweeps. The compute budget supported 10 seeds per cell; some 1–2 sigma effects therefore retain residual seed sensitivity. Pure VAE-family blocks were observed to diverge under iteration-based paper sampling and are excluded *a priori*; we do not claim VAE backbones are unworkable in general, only that they were not stabilized within this study. Finally, all reported gains assume a single architecture is trained per

Table 28: Token-matched validation perplexity (in-domain val set, same harness across variants). The wavelet variant closes the gap to baseline monotonically; the gap-to-`ae_mlp` ratio drops from 1.05 at 250 M tokens to under 1.0 by 1.85 B tokens, so the projection ranking reflects an already-realized ordering rather than a slope-fit artifact.

Tokens	baseline	`ae_mlp`	`tw_root_fc_db3_64_tiered_silu`
250 M	50.28	78.98	68.81
1.00 B	26.42	35.10	33.50
1.85 B	–	–	27.64
2.75 B	19.87	26.52	–
3.0 B	19.82	26.45	24.4 (proj.)

Table 29: Layer-15 AE-LG swap on Llama-3.2-1B with two-phase AE-pretraining and KD fine-tuning. The same block design that pays a $+0.28$-nat plateau from scratch (Section 4.5) recovers below the unmodified original perplexity once the teacher signal is added.

Stage	Validation PPL
Original Llama-3.2-1B (no swap)	19.45
Baseline (PE layer inserted, no KD, no AE pretrain)	42.72
After AE pretrain (teacher-activation reconstruction)	25.34
After KD fine-tune	18.81

dataset/period; ensembling, common in published M4 results, is intentionally outside scope to keep the parameter-count comparison clean.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract's three claims — 60–95% parameter compression over N-BEATS at matched or better accuracy, sub-1M-parameter TrendWavelet models within $\approx 0.5\%$ sMAPE of 19–43M baselines, and transferability to NHiTS and Transformer LLMs — map directly to Tables 5 and 6 (Section 4), Section 4.4, and Section 4.5 / Appendix G.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section H discusses the M4-centric scope of the architectural sweep, the 10-seed-per-cell budget on a single-author setup, residual seed sensitivity on 1–2σ effects, the *a priori* exclusion of pure VAE-family blocks under iteration-based paper sampling, and the no-ensembling convention used to keep the parameter-count comparison clean.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper is empirical; it states no theorems, lemmas, or formal proofs.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Appendix A fixes both paper-sample training protocols (sampling, optimizer, LR schedule, validation cadence, early stopping) verbatim; Appendix B enumerates every `NBeatsNet.__init__` keyword with its paper-sweep value (Table 7) and the per-family sweep grid (Table 8); Appendix B.9 lists the canonical CSV and YAML filenames; the released `lightningnbeats` package implements every block and dataset.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: All experiment YAML configs ship with the submission under `experiments/configs/` (Appendix B.9); the underlying package is published as `lightningnbeats` (PyPI, MIT-licensed) with installation, examples, and a YAML schema documented in the repository; M4, Tourism, Traffic, Weather, and Milk are public benchmarks loaded by the dataset classes in `src/lightningnbeats/data/`; per-run result CSVs are listed in Appendix B.9.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 3 states the three training protocols, datasets, seed budget, and significance test; Appendix A gives the full sampling rule, optimizer, LR schedule, validation cadence, early stopping, and step budget for each paper-sample variant; Appendix B documents block widths, latent and basis dimensions, wavelet families, `active_g`, sweep grids, and per-period recommendations.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Every leaderboard cell reports mean $\pm$ standard deviation over the surviving seeds (10 per cell after applying the strict health filter at the top of Section 4: `diverged`, `sMAPE` > 100, or `best_epoch`$= 0$ with `sMAPE` > 50); pairwise architectural comparisons use the Wilcoxon signed-rank test with Bonferroni correction (Section 3, Appendix B). The variance source is the random seed (initialization plus paper-sample anchor draws).

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix B.10 reports the workstation hardware (2$\times$ NVIDIA RTX 5090 32 GB, AMD Ryzen Threadripper 7970X 32C/64T, 128 GB RAM, CUDA 13.0), the single-GPU-per-run policy, and the per-protocol step or epoch budget that bounds each individual experiment; the YAML configs in `experiments/configs/` fix every per-run training cost.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: The work studies time-series forecasting architectures and small-scale language-model block replacement; it involves no human subjects, no scraped personal data, and no deployed system. All datasets used (M4, Tourism, Traffic, Weather, Milk, FineWeb-Edu) and models studied (SmolLM2, Llama-3.2) are public benchmarks or open-weights releases, used under their respective licenses.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [N/A]

Justification: The contribution is foundational architectural work on parameter-efficient forecasting blocks and Transformer block replacement. There is no direct path to a deployed system, and any dual-use concerns reduce to the generic case of architectural efficiency improvements.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: No high-misuse-risk model or dataset is released. The `lightningnbeats` package contains forecasting blocks and dataset loaders for established public benchmarks; no pretrained large-scale generative model or scraped dataset is distributed.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: M4 [**?**], the original N-BEATS [**?**], NHiTS [**?**], SmolLM2 [**?**], FineWeb-Edu [**?**], Llama-3.2 [**?**], and the `lm-evaluation-harness` [**?**] are all cited in the main text and used under their respective public licenses (CC-BY for M4 and FineWeb-Edu, MIT/Apache for the original N-BEATS and harness code, Llama community license for Llama-3.2).

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The new `lightningnbeats` blocks are documented in `src/lightningnbeats/blocks/blocks.py` with constructor signatures, the package README, and the YAML schema at `experiments/configs/schema.md`; unit tests in `tests/test_blocks.py` cover output shapes and registries. Appendix B is the consolidated hyperparameter reference, including default values, sweep grids, and empirical recommendations.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.

- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper involves no crowdsourcing and no research with human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The paper involves no human subjects research, so IRB review does not apply.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: SmolLM2-135M and Llama-3.2-1B appear in this paper only as experimental subjects for the cross-architecture transfer study in Section 4.5 and Appendix G; they are not part of the core block-design methodology. Any LLM-assisted writing or editing falls within the explicit NeurIPS exemption.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.